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Un ideal es maximal si y solo si el cociente es un cuerpo

Proposición 1. *Sea $I \subsetneq R$ un ideal, entonces*

$$I \text{ es un ideal maximal} \iff {}^{R}\!/_{I} \text{ es un cuerpo.}$$

***Demostración*:** Podemos ver la equivalencia directamente:

$$I \text{ es maximal} \iff I \neq R \wedge \Big(\forall J \subset R \text{ ideal} : I \subset J \subset R \implies I = J \vee J = R\Big)$$

Por el `teo-correspondencia-ideales-cociente/??`, $\forall J \subset R$ ideal que contiene a I, existe un único ideal $\overline{J} \subset {}^{R}\!/_{I}$ tal que $\pi^{-1}(\overline{J}) = J$, donde $\pi : R \to {}^{R}\!/_{I}$ es el morfismo canónico. Así,

$$I \text{ es maximal} \iff {}^{R}\!/_{I} \neq \{0\} \wedge \Big(\forall \overline{J} \subset {}^{R}\!/_{I} \text{ ideal} : \overline{J} = \{0\} \vee \overline{J} = {}^{R}\!/_{I}\Big).$$

Por el `lem-cuerpo-iff-ideales-triviales/??`, esto es equivalente a que ${}^{R}\!/_{I}$ sea un cuerpo. ■

Referenciado en

- `teo-extension-entera-ideal-primo-maximal-iff-maximal`
- `cor-ideal-maximal-imp-primo`

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